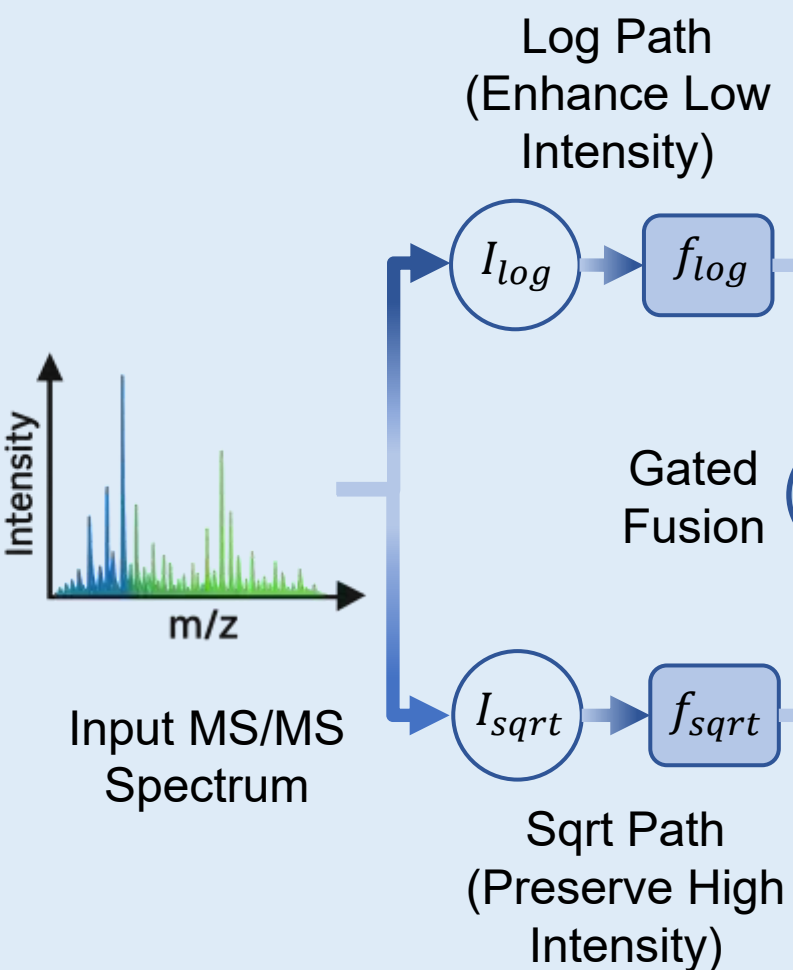
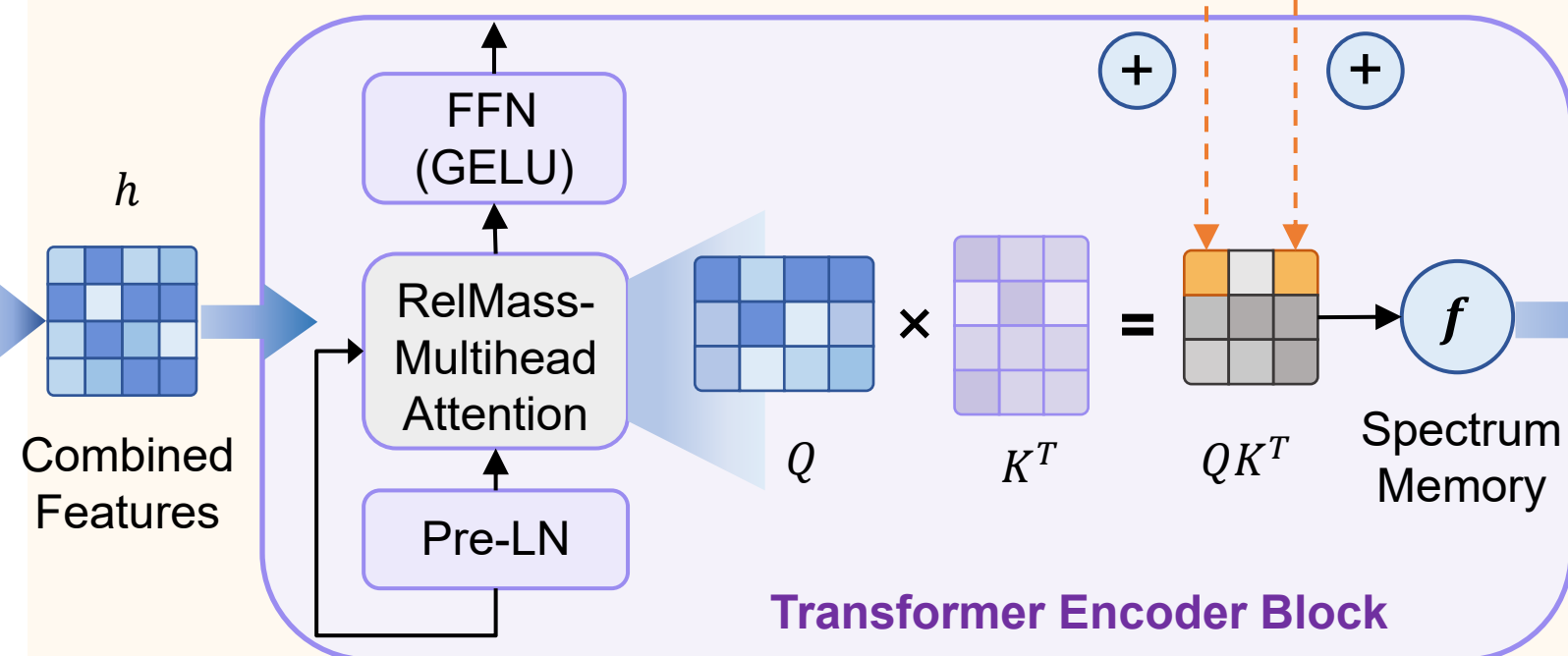
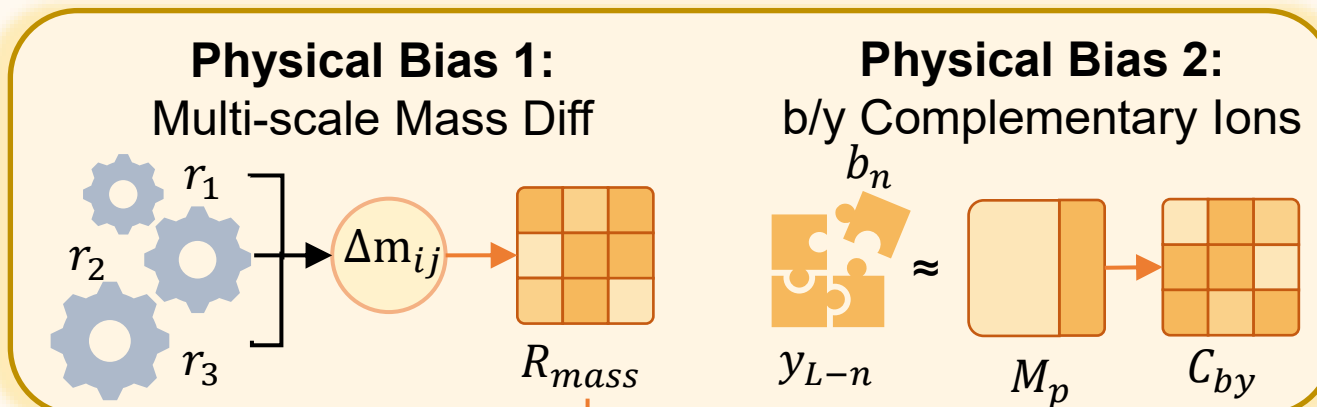


## Module a: Input & Dual-Path Intensity Normalization

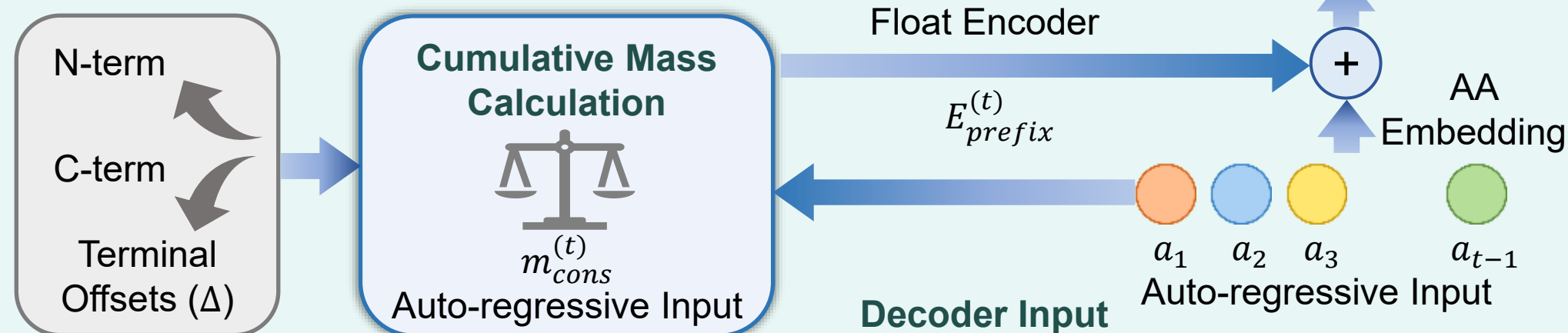
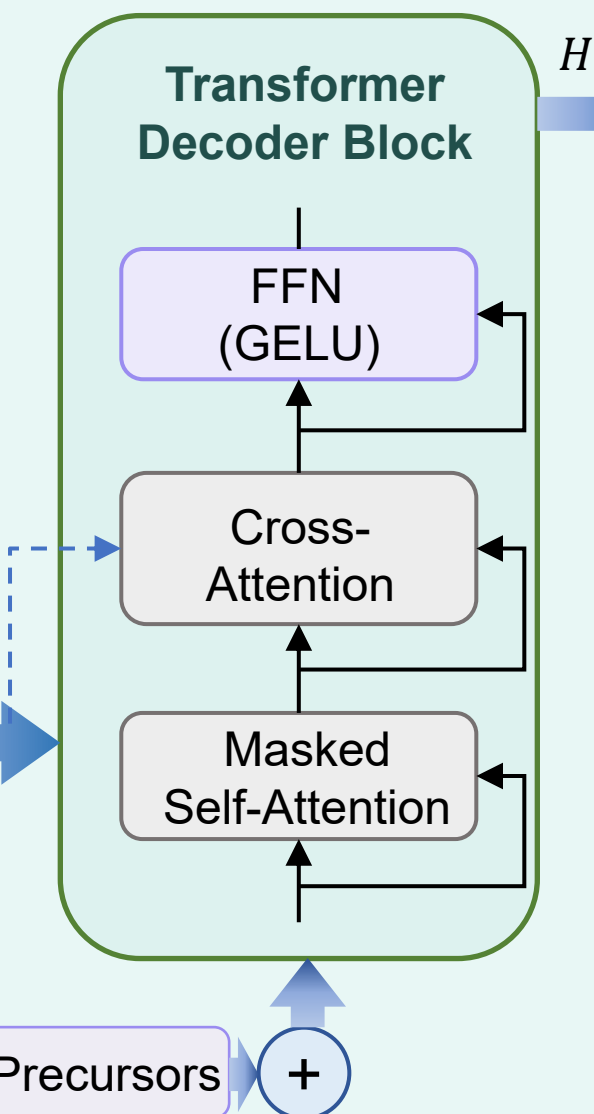


## Module b: Physics-Aware Transformer Encoder

### Physical Constraints

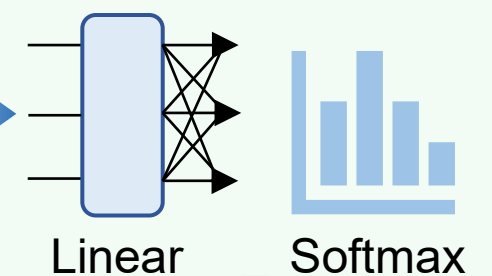


## Module c: Prefix-Mass Guided Transformer Decoder

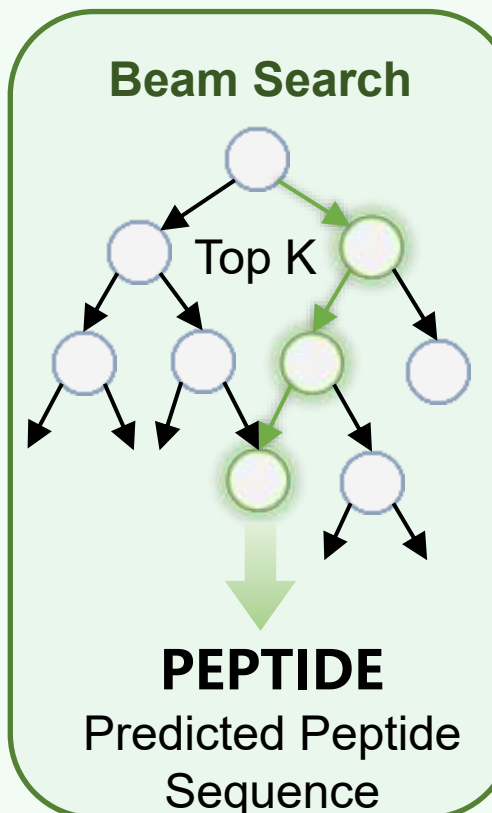


## Module d: Output

### Probability Output



### Search Strategy



### Loss Constraint

